An Optimal Recovery Approach to Interpolation

Ram G. Shenoy, *Member, IEEE*, and Thomas W. Parks, *Fellow, IEEE*

***Abstract*—The design of interpolation filters is reviewed. The Chebyshev and integrated squared-error minimization methods are shown to be optimal for certain classes of input signals. We show that filters designed by integrated squared-error minimization in the frequency domain always interpolate the known samples.**

The interpolation problem is viewed as a problem of recovering the missing samples of a signal. This provides a framework for studying the interpolation problem for more general signal classes, and gives insight into how some of the known results in interpolation filter design arise. For example, it is shown that if the class of input signals is the intersection of a linear variety and a ball in an inner product space, then the optimal interpolation filter is linear and time invariant, even though this is not assumed *a priori*. We also show that the problem of estimating missing samples (optimal recovery), is equivalent to a problem of approximating the representer of a missing sample by a linear combination of the representers of the known samples. This approximation problem is shown to be the same as the minimization of an integrated squared error in the frequency domain, thus demonstrating the connection between optimal recovery, and a classical method of interpolation filter design.

I. Introduction

INTERPOLATION filters are important for several signal processing applications: for example, they are used to upsample in multirate processing. This article is about the design of such filters. The results we describe are not new: however, we feel that it is worthwhile to collect them together. Furthermore, it appears that some of these results are not being exploited in the signal processing community, and it is hoped that this presentation will provide a new perspective that will lead to more widespread use of these ideas.

The interpolation problem is generally viewed from the perspective of filter design. In the filter design problem, one tries to make the frequency response of the actual filter approximate to a prescribed desired frequency response as closely as possible, taking into account prior knowledge about the inputs. The error frequency response is the difference of the desired frequency response and the actual frequency response, and an approximation criterion is selected to make the error frequency response as close to zero as possible. Commonly used criteria are the minimization of the L^∞, or Chebyshev, norm (maximum amplitude), and of the L^2 norm (integrated squared amplitude) of the error frequency response. We show that these criteria are appropriate when the input to the interpolation filter belongs to a filter class: the set of possible outputs from a hypothetical linear time-invariant filter, with bounded energy inputs. The minimization of a Chebyshev norm in the frequency domain is shown to minimize a worst case integrated squared error in the time domain, over all signals in the filter class: the filters so designed do not interpolate the known samples. On the other hand, we demonstrate that minimization of the L^2 norm in the frequency domain minimizes a worst case pointwise error in the time domain, over all signals in the filter class. We show that such a filter does interpolate the known samples.

Alternatively, the design of an interpolation filter may be viewed as a problem of estimating missing samples of a signal. This latter problem can be examined using the theory of optimal recovery. The theory of optimal recovery provides a broader setting, which not only illuminates some of the results on interpolation filters, such as in [1], but also indicates how to analyze the interpolation problem when the signal inputs belong to classes other than the filter classes that we treat here [2].

We look at optimal recovery of a single unknown time sample from known samples, and from knowledge that the signal belongs to a filter class. The techniques of Golomb and Weinberger [3] give the best estimate of a sample at a time index n: it is the value, at n, of the signal which goes through the known samples, and which minimizes a weighted l_2-norm. From Golomb and Weinberger [3], the estimate is best in the sense that it is the midway between the two extreme values that the sample may take. Equivalently, we show that from the optimal recovery view, the estimate is best because it minimizes a worst case error, over signals in the filter class. We remark that the form of the filter class assumed implies that minimum norm signal is also the best pointwise estimate over all signals in the filter class: the minimum norm signal minimizes a worst case l_∞ error over all signals in the class. It follows that the filter which generates the minimum norm signal from the known samples is the best l_∞-interpolation filter.

The Golomb–Weinberger theory of estimating linear functionals is related to the theory of splines. Finding a minimum norm signal which goes through the known samples may equivalently be regarded as finding a spline

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R. G. Shenoy was with the School of Electrical Engineering, Cornell University, Ithaca, NY 14853. He is now with Schlumberger-Doll Research, Ridgefield, CT 06877.

T. W. Parks is with the School of Electrical Engineering, Cornell University, Ithaca, NY 14853.

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function which satisfies a generalized "smoothness" criterion. This relationship has already been observed in [4] and [5], and we shall discuss this briefly, when we review minimum norm extrapolation.

After stating the interpolation problem, we define a filter class as a ball in an inner product space, and apply some standard results on inner product spaces to filter classes. We then outline the filter design problem and describe the work of Oetken *et al.* [1]. Subsequently, we review the theory of optimal recovery, which is more fully described in [6], and examine the interpolation problem within an optimal recovery context. Using the observations of Golomb and Weinberger [3], we show that the interpolation problem may be reduced to studying a hypercircle in an inner product space. We introduce the notion of the Chebyshev center of a set, and remark that the solution to the interpolation problem, from the optimal recovery viewpoint, is to find the Chebyshev center of the hypercircle. The interpolation filter is then the operator which transforms the vector of known samples into the center of the hypercircle. Since the geometry of the hypercircle is particularly simple, we can deduce some auxiliary results such as the linearity and time invariance of the interpolation filter. Then we show that the estimation of an unknown sample is the same as the problem of approximating the representer of the unknown sample by a linear combination of the representers of the known samples. Hence we can deduce that the interpolation problem is equivalent to minimizing the L^2 norm of the error frequency response, from the filter design point of view.

II. Notation

The set of real numbers is $\boldsymbol{R}$ and the set of integers is $\boldsymbol{Z}$. The space of discrete-time finite energy signals is $l_2(\boldsymbol{Z})$, commonly shortened to l_2. If x is in l_2, then its Fourier transform is $X(\omega)$, $\omega \in [-\pi, \pi)$ and its value at time n is x_n. By Parseval's theorem, $X(\omega)$ belongs to $L^2([-\pi, \pi))$, the space of square integrable functions on $[-\pi, \pi)$. If $\mathfrak{X}$ is an inner product space, the inner product and norm associated with it are $(\cdot, \cdot)_{\mathfrak{X}}$ and $\|\cdot\|_{\mathfrak{X}}$ respectively. In the case of l_2 or $L^2([-\pi, \pi))$, the norm is denoted by $\|\cdot\|_2$: the space should be apparent from the context.

We will be discussing linear time-invariant filters and shall use the following convention. The impulse response of a filter will be given in lower case, e.g., h_n, and the frequency response, being the Fourier transform of the impulse response, will be in upper case, i.e., $H(\omega)$. The action of this filter upon a signal, $x \in l_2$, is that of a linear operator acting upon x, and we denote this linear operator by bold-face upper case $\boldsymbol{H}$. We shall refer to the filter and the corresponding operator interchangeably, e.g., we shall talk about the filter $\boldsymbol{H}$.

Recall that a subset B of a linear space $\mathfrak{X}$ is convex if the line segment joining two elements of B, x, and y, is contained in B:

$$x, y \in B \Rightarrow \lambda x + (1 - \lambda)y \in B \qquad 0 \le \lambda \le 1.$$

The set B is said to be balanced if it is symmetric around the origin: x in B implies that $-x$ is in B.

III. Problem Statement

Consider a sequence $y = \{y_k\}$, which is the result of decimating an unknown discrete-time signal $x = \{x_k\}$

$$y_k = \begin{cases} x_k, & \forall k = \lambda r \quad (1) \\ 0, & \forall k \neq \lambda r \quad (2) \end{cases}$$

where $r > 1$ is an integer, $\lambda = \cdots, -1, 0, +1, \cdots$.

We wish to design a time-varying interpolation filter using the available prior knowledge about the unknown sequence $\{x_k\}$ so that its output sequence approximates to $\{x_k\}$ as closely as possible. We shall explain the phrase "as closely as possible" subsequently: briefly, we shall choose an estimate which will minimize the norm of a worst case error. The prior knowledge that we have about $\{x_k\}$ is that it belongs to a filter class: it is a possible output of a linear time-invariant filter with bounded energy inputs.

IV. Filter Classes

A. Signal Space and Weighted Inner Products

We define filter classes to continue developing the interpolation problem. Filter classes are used to model prior knowledge about the input signals to the interpolation filter.

Let the subspace $\mathfrak{X}$ of l_2 be defined as

$$\mathfrak{X} = \{x: X(\omega) = H(\omega)U(\omega),\ u \in l_2\}. \qquad (3)$$

In (3), $H(\omega)$ is the Fourier transform of an absolutely summable sequence, $\{h_n\} \in l_1$. The set $\mathfrak{X}$ is the set of all possible outputs from a hypothetical filter, with impulse response $\{h_n\}$, with finite energy inputs. The frequency response of this filter is supposed to incorporate prior knowledge about the signals being interpolated. The bounded linear operator $\boldsymbol{H}$: $l_2 \to \mathfrak{X}$, denotes the action of the filter with frequency response $H(\omega)$ and impulse response $\{h_n\}$. The norm of $\boldsymbol{H}$ as an operator from l_2 onto $\mathfrak{X}$ with the l_2 norm, $\|\boldsymbol{H}\|$, is $\|H(\omega)\|_\infty$ [7], and we take it to be unity, without loss of generality.

However, instead of using the l_2 norm for $\mathfrak{X}$, we introduce a weighted l_2-norm based upon the operator $\boldsymbol{H}$. First recall the definition of the "pseudoinverse" of $\boldsymbol{H}$, which we shall denote by $\boldsymbol{H}^{-1}$. The null space (kernel) of $\boldsymbol{H}$, N, is the set of signals that are annihilated by $\boldsymbol{H}$:

$$N = \{u \in l_2: \boldsymbol{H}u = 0\}.$$

Since $\boldsymbol{H}$ is a bounded operator, N is a closed subspace. The projection, $P_N u$, of an l_2 signal, u, onto N, is the signal in N which minimizes the distance between N and u

$$\|u - P_N u\|_2 = \inf_{u_0 \in N} \|u - u_0\|_2.$$

Note that if u is in N, then $P_N u = u$. If $\boldsymbol{H}u_1 = \boldsymbol{H}u_2 = x$ for two signals u_1 and u_2 in l_2, then $u_1 - P_N u_1 = u_2 - P_N u_2$. For since $\boldsymbol{H}$ annihilates $u_1 - u_2$, N contains $u_1 - u_2$, and $u_1 - u_2 = P_N(u_1 - u_2)$. The pseudoinverse of $\boldsymbol{H}$ is then defined as $\boldsymbol{H}^{-1}$: $\boldsymbol{H}u \to u - P_N u$. This defines $\boldsymbol{H}^{-1}$ on the range of $\boldsymbol{H}$, ran $(\boldsymbol{H})$. We define $\boldsymbol{H}^{-1}x = 0$ for x in the orthogonal complement of ran $(\boldsymbol{H})$ in l_2. This is of no

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concern to us, however, since the signals in $\mathfrak{X}$ all lie in the range of $\boldsymbol{H}$. When $\boldsymbol{H}$ is invertible, $\boldsymbol{H}^{-1}$ is just the normal inverse.

For a signal x in $\mathfrak{X}$, $\boldsymbol{H}^{-1}x$ is the signal of minimum norm in l_2 such that $\boldsymbol{H}\boldsymbol{H}^{-1}x = x$:

$$\|\boldsymbol{H}^{-1}x\|_2 = \inf_{\boldsymbol{H}u = x} \|u\|_2.$$

In frequency domain terms,

$$H^{-1}(\omega) = \begin{cases} \dfrac{1}{H(\omega)}, & H(\omega) \neq 0 \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

In (4), $H^{-1}(\omega)$ is the frequency response of the filter which corresponds to the linear operator $\boldsymbol{H}^{-1}$. The pseudoinverse gives the following alternative form for $\mathfrak{X}$:

$$\mathfrak{X} = \left\{ x \in l_2 \colon \int_{-\pi}^{\pi} |H^{-1}(\omega)|^2 |X(\omega)|^2 \, d\omega < \infty, \right.$$
$$\left. X(\omega) = 1_D(\omega) X(\omega) \right\}. \tag{5}$$

In (5), D is the support of $H(\omega)$, and $1_D(\omega)$ is the set indicator function of D: it takes the value 1 for ω in the D, and 0 otherwise. We note the following useful relationship between $1_D(\omega)$ and $H(\omega)$ for future reference:

$$1_D(\omega) = H^{-1}(\omega)H(\omega) = |H^{-1}(\omega)|^2 |H(\omega)|^2.$$

The characterization in (5) leads to the definition of a weighted inner product and norm on $\mathfrak{X}$:

$$(x, y)_{\mathfrak{X}} = (\boldsymbol{H}^{-1}x, \boldsymbol{H}^{-1}y)_2$$
$$\|x\|_{\mathfrak{X}} = \|\boldsymbol{H}^{-1}x\|_2. \tag{6}$$

Writing this set of equations in the frequency domain, we get

$$(x, y)_{\mathfrak{X}} = \int_{-\pi}^{\pi} |H^{-1}(\omega)|^2 X(\omega) \overline{Y}(\omega) \, d\omega$$
$$\|x\|_{\mathfrak{X}} = \int_{-\pi}^{\pi} |H^{-1}(\omega)|^2 |X(\omega)|^2 \, d\omega.$$

B. Samples as Linear Functionals

The value of a signal at time index n, x_n, is a bounded linear functional of x in the $\mathfrak{X}$ norm. This can be seen by manipulating the inverse Fourier transform relationship:

$$x_n = \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} \, d\omega = \int_{-\pi}^{\pi} 1_D(\omega) X(\omega) e^{j\omega n} \, d\omega$$
$$= \int_{-\pi}^{\pi} |H^{-1}(\omega)|^2 |H(\omega)|^2 e^{j\omega n} X(\omega) \, d\omega \qquad x \in \mathfrak{X}.$$

The sample value at the nth instant, x_n, is the $\mathfrak{X}$ inner product of x with a shifted version of r, the inverse Fourier transform of $|H(\omega)|^2$. This is an example of the Riesz representation theorem: to every bounded linear functional f on a Hilbert space, is associated a unique vector in the Hilbert space, the representer ϕ_f. The action of f on an element x is equivalent to taking the inner product of x with the representer

$$f(x) = (\phi_f, x).$$

The representer of the sample value at time index n, x_n, is $\{r_{n-\nu}\}_{\nu=-\infty}^{\infty}$, where $\{r_\nu\}$ is the inverse Fourier transform of $|H(\omega)|^2$. Denote this representer by ϕ_n:

$$x_n = (\phi_n, x)_{\mathfrak{X}}.$$

The set of signals orthogonal to ϕ_n is the set of signals which are zero at time index n: we call this subspace F_n.

C. Definition of Filter Classes

We define a filter class K as the image under $\boldsymbol{H}$ of the unit ball in l_2:

$$K = \{x \colon X(\omega) = H(\omega)U(\omega), \, u \in l_2, \, \|u\|_2 \leq 1\}. \tag{7}$$

With the norm and inner product defined on $\mathfrak{X}$, K is the unit ball in $\mathfrak{X}$. To see this pick $z \in \{u \in l_2 \colon \|u\|_2 \leq 1\}$. Then

$$\|\boldsymbol{H}u\|_{\mathfrak{X}} = \|\boldsymbol{H}(u - P_N u)\|_{\mathfrak{X}}$$
$$= \|\boldsymbol{H}^{-1}\boldsymbol{H}(u - P_N u)\|_2$$
$$= \|u - P_N u\|_2 \leq \|u\|_2 \leq 1.$$

The first inequality follows since, by definition, $u - P_N u$ is the minimum norm vector such that $\boldsymbol{H}(u - P_N u) = \boldsymbol{H}u$. It is clear that the upper bound on the $\mathfrak{X}$-norm can be achieved: pick $u' = (u - P_N u)/\|u - P_N u\|_2$.

Having defined filter classes, we can now describe the interpolation problem from the viewpoint of filter design.

V. The Filter Design Problem

Fig. 1 shows the elements of the filter design problem. A discrete time signal x is to be filtered by a linear time-invariant operator. A desired frequency response $D(\omega)$ is prescribed. The frequency response of the filter that we actually use is $A(\omega)$. We analyze the error filter $\boldsymbol{E}$ with frequency response $E(\omega) = D(\omega) - A(\omega)$. We suppose that any *a priori* information about x is described by saying that x belongs to the set of possible outputs of the hypothetical filter, $\boldsymbol{H}$, with l_2 inputs of unit energy. There are several optimization criteria that one may choose to make $A(\omega)$ approximate to $D(\omega)$. One example is Chebyshev error minimization in the frequency domain: we minimize the L_∞ norm of the error frequency response, $E(\omega) = D(\omega) = A(\omega)$. This is appropriate for minimizing the maximum output energy of the error filter, as we shall now demonstrate.

Suppose that the inputs to the filter being designed have frequency support contained in the set $\mathfrak{D}$, which is the union of disjoint intervals, as shown in Fig. 2. Equivalently, we may think of these input signals as belonging to the set of possible outputs of a linear time-invariant filter $\boldsymbol{H}$. The frequency response of $\boldsymbol{H}$ is the set indicator function of $\mathfrak{D}$: $H(\omega) = 1_{\mathfrak{D}}(\omega)$. The desired frequency response consists of certain pass bands and stopbands on $\mathfrak{D}$ and is unspecified off $\mathfrak{D}$. The maximum output energy of the error filter, over all the possible inputs, in the fre-

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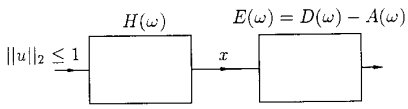

Fig. 1. The filter design problem.

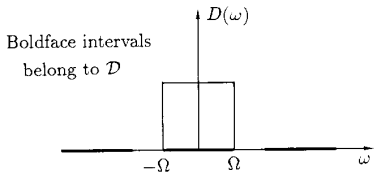

Fig. 2. Desired frequency response.

quency bands contained in $\mathfrak{D}$, is

$$\sup_{\|u\|_2 \le 1} \|\boldsymbol{EHu}\|_2 = \sup_{\omega \in \mathfrak{D}} |E(\omega)| = \|E(\omega)\|_\infty. \tag{8}$$

The first equality in (8) is a standard result in analysis, see, e.g., [7]. In order to minimize the maximum output energy from the error filter, over all signals in the filter class, one has to minimize the maximum amplitude (L_∞ norm) of the error frequency response over $\mathfrak{D}$. The filter design problem is reduced to an L_∞-approximation problem on the domain $\mathfrak{D}$.

Alternatively, suppose that instead of minimizing the output energy of the error filter, it is desired to minimize the output of the error filter, at a single time-index n over all signals in the filter class. The output of the error filter at time n is

$$(\boldsymbol{Ex})_n = \int_{\mathfrak{D}} E(\omega) e^{j\omega n} X(\omega)\, d\omega.$$

Equivalently, we can view $(\boldsymbol{Ex})_n$ as a linear functional of the signal x, whose representer is the inverse Fourier transform of $1_{\mathfrak{D}}(\omega)E(\omega)e^{j\omega n}$: this is a shifted version of the impulse response of $\boldsymbol{E}$, filtered by $\boldsymbol{H}$. Using the Schwarz inequality, the maximum value of $(\boldsymbol{Ex})_n$ over all possible inputs is

$$\sup_{\|u\|_2 \le 1} |(\boldsymbol{EHu})_n| = \left(\int_{\mathfrak{D}} |E(\omega)|^2\, d\omega \right)^{1/2}. \tag{9}$$

Since the right hand side of (9) is independent of n, minimizing the output spread of the error filter at a particular time index, n, minimizes the output spread for all time indices. Hence, minimizing this integrated squared error actually minimizes $\sup_n \sup_{\|u\|_2 \le 1} |(\boldsymbol{EHu})_n|$. Once again, the filter design problem is reduced to an approximation problem: this time it is an L_2 approximation on $\mathfrak{D}$. Note that this frequency domain integrated squared error minimization actually minimizes the maximum l_∞ norm of the error filter output. This follows from the equality

$$\sup_n \sup_{\|u\|_2 \le 1} |(\boldsymbol{EHu})_n| = \sup_{\|u\|_2 \le 1} \sup_n |(\boldsymbol{EHu})_n| = \sup_{\|u\|_2 \le 1} \|\boldsymbol{EHu}\|_\infty.$$

Fig. 3 shows a filter design problem which we will relate to the band-limited interpolation problem. We introduce the sampling element after the filter $\boldsymbol{H}$: this lets every rth sample pass and zeros out the others. The filter $\boldsymbol{H}$ is an ideal low-pass filter, its frequency response being the indicator function of an interval, $\mathfrak{D} = [-\Omega, \Omega]$, centered at the origin. This is also the desired frequency response $D(\omega)$. Henceforth we shall fix $\mathfrak{D} = [-\Omega, \Omega]$. We suppose the decimation rate is such that the decimated signal is still oversampled, i.e., $|\Omega| < \pi/r$. We wish to design $A(\omega)$ to minimize the output spread at a single time-index n. The sampled signal may be regarded as the product, in the time domain, of the signal x, and a sequence of period r, of unit delta functions. It follows that the spectrum of the sampled signal $X_s(\omega)$, is the convolution of $X(\omega)$ and the Fourier transform of this sequence:

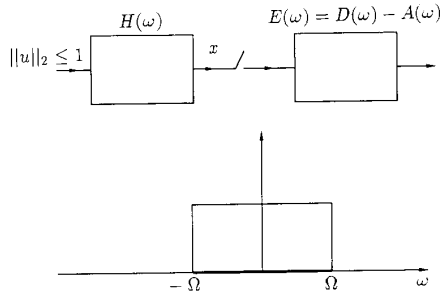

Fig. 3. Filter design for band-limited interpolation.

$$X_s(\omega) = \frac{1}{r} \int_{-\pi}^{\pi} X(\eta) \sum_{l:|2\pi l/r| \le \pi} \delta\left(\omega - \eta - \frac{2\pi l}{r}\right) d\eta. \tag{10}$$

The output of the error filter at time n is

$$(\boldsymbol{Ex})_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} - X_s(\omega) \sum_k a_k e^{j\omega(n-k)}\, d\omega$$
$$x = \boldsymbol{Hu},\ \|u\|_2 \le 1. \tag{11}$$

Substituting (10) in (11), and interchanging the order of integration, we get

$$(\boldsymbol{Ex})_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) \cdot \left(e^{j\omega n} - \sum_k a_k e^{j\omega(n-k)} \frac{1 \sin(\pi(n-k))}{r \sin\left(\frac{\pi}{r}(n-k)\right)} \right) d\omega. \tag{12}$$

We may again use the Schwarz inequality to maximize (12) over all u,

$$\sup_{\|u\|_2 \le 1} |(\boldsymbol{EHu})_n| = \left(\int_{[-\Omega,\Omega]} \left| e^{j\omega n} - \sum_k a_k e^{j\omega(n-k)} \cdot \frac{1}{r} \frac{\sin(\pi(n-k))}{\sin(\pi(n-k)/r)} \right|^2 d\omega \right)^{1/2}. \tag{13}$$

In order to minimize the maximum output spread, we must choose the coefficients $\{a_k\}$ to minimize (13). At this stage, it is not apparent that the best such filter will be an

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interpolating filter, nor that it minimizes the output spread for all time indices.

To minimize (13), we observe that the quotient of sines term is of the form

$$\frac{1}{r}\frac{\sin(\pi(n-k))}{\sin(\pi(n-k)/r)} = \begin{cases} 1, & \text{if } n-k \equiv 0 \bmod (r) \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

This has two consequences. If $n \equiv \rho \bmod (r)$, where $\rho = 0, \cdots, r-1$, then minimizing (13) for n is the same as minimizing (13) for ρ. For fixed ρ, the solution depends only upon $\{a_{\lambda r+\rho}\}_{\lambda=-\infty}^{\infty} i$ and is independent of a_k for $k \not\equiv \rho \bmod (r)$. Hence we need solve r independent minimization problems. For each ρ, we obtain $\{a_{\lambda r+\rho}\}_{\lambda=-\infty}^{\infty}$. The resulting filter minimizes the output spread for all time indices. Furthermore, by minimizing (13) for the case $\rho = 0$, we see that the filter is an interpolation filter. The method of solving r independent subproblems to get an interpolation filter was used in Oetken *et al.* [1]. We remark that, after substituting (14) in (13) it has the form of approximating the representer of the nth time index by a linear combination of the representers of the samples on the time-indices $\{\lambda r\}_{\lambda=-\infty}^{\infty}$:

$$\sup_{\|u\|_2 \le 1} |(\boldsymbol{EHu})_n| = \left(\int_{[-\Omega,\Omega]} \left| e^{j\omega n} - \sum_{\lambda=-\infty}^{\infty} a_{\lambda r+\rho} e^{j\omega(n-\lambda r-\rho)} \right|^2 d\omega\right)^{1/2}$$
$$n \equiv \rho \bmod (r). \tag{15}$$

We may use the orthogonality principle to obtain a system of equations from (15):

$$\int_{[-\Omega,\Omega]} e^{j\omega n}\, d\omega = \int_{[-\Omega,\Omega]} \sum_{\lambda=-\infty}^{\infty} a_{\lambda r+\rho} e^{j\omega(n-\lambda r-\rho)}\, d\omega$$
$$n \equiv \rho \bmod (r). \tag{16}$$

We show that the filter design problem shown in Fig. 4 is actually equivalent to the one shown in Fig. 3. The frequency response of the filter $\boldsymbol{H}_s$ is a "sampled" version of the frequency response of $\boldsymbol{H}$ in Fig. 3:

$$H_s(\omega) = \sum_{l:|2\pi l/r| \le \pi} H\left(\omega + \frac{2\pi l}{r}\right) = \sum_{l:|2\pi l/r| \le \pi} 1_{\mathfrak{D}}\left(\omega + \frac{2\pi l}{r}\right).$$

The support of $H_s(\omega)$ is $\mathfrak{D}_s$. The desired frequency response is again the indicator function of the interval $\mathfrak{D} = [-\Omega, \Omega]$ and we choose filter coefficients a_k to minimize the output spread at time n. Using (9), the best such filter minimizes an integrated squared error in the frequency domain:

$$\sup_{\|u\|_2 \le 1} |(\boldsymbol{EH}_s\boldsymbol{u})_n| = \left(\int_{\mathfrak{D}_s} \left| D(\omega) - \sum_k e^{-j\omega k} \right|^2 d\omega\right). \tag{17}$$

To minimize (17), we apply the orthogonality principle

$$\int_{\mathfrak{D}_s} D(\omega) e^{j\omega n}\, d\omega = \int_{\mathfrak{D}_s} \sum_k a_k e^{j\omega(n-k)}\, d\omega. \tag{18}$$

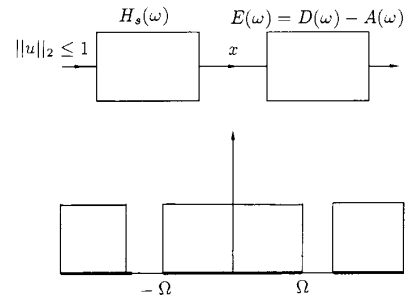

Fig. 4. Second filter design problem related to band-limited interpolation.

We can simplify (18) by noting that the complex exponentials have a curious orthogonality property on the domain $\mathfrak{D}_s$:

$$\frac{1}{2\pi}\int_{\mathfrak{D}_s} e^{j\omega(n-m)} = \begin{cases} \dfrac{\Omega}{\pi}\dfrac{\sin(\Omega(n-m))}{\Omega(n-m)}, & \text{if } n-k \equiv 0 \bmod (r) \\ 0, & \text{otherwise.} \end{cases} \tag{19}$$

This property becomes self-evident, once we observe that the left-hand side of (19) is the inverse Fourier transform of a sinc function of bandwidth Ω, sampled every rth instant. Using (19), and noting that $D(\omega)$ is the indicator function of the interval $[-\Omega, \Omega]$, we can simplify (18) to

$$\int_{[-\Omega,\Omega]} e^{j\omega n}\, d\omega = \int_{[-\Omega,\Omega]} \sum_{\lambda=-\infty}^{\infty} a_{\lambda r+\rho} e^{j\omega(n-\lambda r-\rho)}\, d\omega$$
$$n \equiv \rho \bmod (r). \tag{20}$$

Since (16) and (20) are identical, it follows that the filter minimizing the integrated squared error in Fig. 4 is actually the interpolating filter in Fig. 3.

VI. Optimal Recovery

We describe the optimal recovery theory of Michelli and Rivlin [8], with a view to applying it to the interpolation problem. Fig. 5 shows the elements of optimal recovery. Let $\boldsymbol{U}: \mathfrak{X} \to \mathfrak{Z}$ be a linear operator acting from a linear space $\mathfrak{X}$ to a normed linear space $\mathfrak{Z}$. The aim is to approximate $\boldsymbol{U}x$, for $x \in K$, where K is a convex balanced subset of $\mathfrak{X}$. The available knowledge about x is given by $y = \boldsymbol{I}x$, where $\boldsymbol{I}: \mathfrak{X} \to \mathfrak{Y}$ is another linear operator acting from $\mathfrak{X}$ into a second normed linear space $\mathfrak{Y}$. To recover $\boldsymbol{U}x$, we apply a (not necessarily linear) map $\boldsymbol{A}: \mathfrak{Y} \to \mathfrak{Z}$ to y.

Within the context of signal processing, $\mathfrak{X}$ may be regarded as a signal space incorporating prior knowledge, modeling those signals relevant to the problem. For the stated interpolation problem, we take $\mathfrak{X}$ to be the space of output of a linear time-invariant filter as in (3). The $\mathfrak{X}$-norm is the weighted l_2-norm based on the hypothetical

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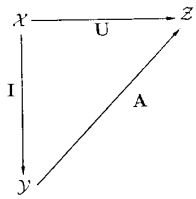

Fig. 5. The elements of optimal recovery.

filter $\boldsymbol{H}$, and K is the unit ball in this norm. The space $\mathcal{Z}$ models the parameters that we are interested in estimating. For example, if a missing sample is to be estimated, then $\mathcal{Z}$ is $\boldsymbol{R}$. More generally if L samples are to be estimated, then $\mathcal{Z}$ is the L-dimensional vector space $\boldsymbol{R}^L$. If the whole signal is to be recovered we take $\mathcal{Z}$ to be $\mathfrak{X}$, and, in this case, $\boldsymbol{U}$ is then the identity operator.

The "information" operator $\boldsymbol{I}$ maps the unknown signal x into the space $\mathcal{Y}$ which models the parameters of x that are known. For our problem, $\boldsymbol{I}$ samples x at infinitely many uniformly spaced points. Let $T = \{\lambda r: \lambda \in \boldsymbol{Z}\}$ be the subset of $\boldsymbol{Z}$ on which we sample x. Then the range of $\boldsymbol{I}$ is contained in $l_2(T)$. We regard $l_2(T)$ as a subspace embedded in l_2 and take $\mathcal{Y}$ to be l_2. If y is in $l_2(T)$, then $\{y_n\}$ denotes the components of the vector y: y_n can be nonzero only for n in T, and it is zero otherwise. Fig. 6 shows the interpolation of a single missing sample posed as an optimal recovery problem.

The operator $\boldsymbol{A}$, when applied to y, gives an estimate for Ux. The data vector y determines a certain section of the set K:

$$C(y) = \{x: \boldsymbol{I}x = y, x \in K\}.$$

We shall call $C(y)$ the hypercircle determined by y, anticipating the geometric picture we shall shortly provide. Following Michelli and Rivlin [8], we will adopt the "Chebyshev center" of $\boldsymbol{U}C(y)$, a notion to be defined, as the solution of the optimal recovery problem. The motivation for using this is that the Chebyshev center minimizes a worst case error norm in $\mathcal{Z}$. So the solution to the recovery problem is

$$\boldsymbol{A}y = c(\boldsymbol{U}C(y))$$

where $c(\boldsymbol{U}C(y))$ is the Chebyshev center of $\boldsymbol{U}C(y)$.

Fig. 7 is a useful geometric picture of $\mathfrak{X}$. The filter class K is represented by the ellipsoid. The known samples determine a hyperplane in $\mathfrak{X}$. The intersection of the hyperplane and ball is a hypercircle in $\mathfrak{X}$. This intersection depends upon the known sample y: we call it $C(y)$. Formally,

$$C(y) = \{x \in \mathfrak{X}: \boldsymbol{I}x = y, \|x\|_{\mathfrak{X}} \le 1\}. \tag{21}$$

The image of $C(y)$ under $\boldsymbol{U}$ is the range of values that $\boldsymbol{U}x$ can take, and the optimal recovery problem is to select a value in $\mathcal{Z}$ which is a best approximation over all $\boldsymbol{U}x$ in $\boldsymbol{U}C(y)$.

We can regard interpolation as finding an estimate (x_n) for each missing sample x_n. Having estimated all the missing samples, we can view the collection of sample estimates $\{(x_n)\}$ together with the known samples, as an estimate for x, $\hat{x}$. However, it does not necessarily follow that $\hat{x}$ will lie in the filter class: this will be shown to be true because the set K is convex and balanced, and $\mathfrak{X}$ is an inner product space. Hence, the collection of sample estimates $\hat{x}$ is a "pointwise" best estimate for x: it will be shown to minimize a worst case l_∞ error norm. Equivalently, we shall show that $\hat{x}$ is the answer to the following optimal recovery problem. We set $\mathcal{Z}$ equal to X with the l_∞ norm and $\boldsymbol{U}$ is the identity operator, mapping $\mathfrak{X}$, with a norm associated with an inner product, to $\mathfrak{X}$ with the l_∞ norm. Then finding the best estimate for x in the l_∞ norm is equivalent to estimating each missing sample individually. To show this, we will introduce Chebyshev centers and discuss some of their properties.

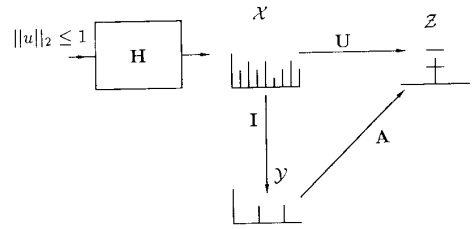

Fig. 6. Estimation of a missing sample.

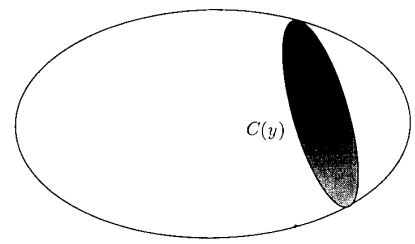

Fig. 7. Geometric diagram of filter class.

VII. Chebyshev Centers

A. Minimum Norm Extrapolation

Before discussing Chebyshev centers, we note that one technique for extrapolation is to take the minimum energy signal, which, when filtered by $\boldsymbol{H}$, goes through the known samples. We minimize the l_2 norm of the signal, which, after filtering by $\boldsymbol{H}$, agrees with the known samples. If y is the vector of known samples, then the minimum norm signal is $(\boldsymbol{IH})^{-1}y$, where $(\boldsymbol{IH})^{-1}$ is the pseudoinverse of the operator $\boldsymbol{IH}$: $l_2 \to \mathcal{Y}$. Since $\|x\|_{\mathfrak{X}} = \|\boldsymbol{H}^{-1}x\|_2$, we remark that the above problem is equivalent to finding the signal, x in $\mathfrak{X}$, which goes through the known sample points and minimizes the $\mathfrak{X}$-norm:

$$\|(\boldsymbol{IH})^{-1}y\|_2 = \inf_{\boldsymbol{IH}u = y} \|u\|_2 = \inf_{\boldsymbol{IH}u = y} \|\boldsymbol{H}u\|_{\mathfrak{X}} = \|\boldsymbol{I}^{-1}y\|_{\mathfrak{X}}. \tag{22}$$

In (22), $\boldsymbol{I}^{-1}$: $\mathcal{Y} \to \mathfrak{X}$ is the pseudoinverse of $\boldsymbol{I}$. It will be

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shown that $I^{-1}y$, an element of $\mathfrak{X}$, is the "Chebyshev center" of the hypercircle $C(y)$.

We should mention the relation between the theory of splines and (22). A fuller discussion may be found in [9, p. 7]; the original papers are [4] and [5]. The element $IH^{-1}y$ may be thought of as a generalized spline, minimizing the "smoothness" constraint defined by the second equality in (22). The filter, H may be viewed as a smoothing operator, and (22) expresses the minimum norm property that characterizes the generalized spline, where the norm is associated to the smoothing operator.

B. Definition of Chebyshev Centers

The Chebyshev center of a set is a best approximation to the elements in that set. A signal z_0 is a Chebyshev center for a subset M of $\mathfrak{X}$ if

$$\sup_{a\in M} \|z_0 - a\|_{\mathfrak{X}} = \inf_{z\in\mathfrak{X}} \sup_{a\in M} \|z - a\|_{\mathfrak{X}}.$$

A Chebyshev center minimizes the worst deviation in the $\mathfrak{X}$-norm over all elements in M. For the sets that we consider, the Chebyshev center will be unique: we denote the Chebyshev center of a set M by $c(M)$.

The zero element is a Chebyshev center for a balanced set B. This is intuitively reasonable, and the proof is straightforward. For any fixed nonzero element, z in $\mathfrak{X}$, the following relationship holds, for balanced B:

$$\sup_{a\in B} \|z - a\|_{\mathfrak{X}} = \sup_{a\in B} \|z + a\|_{\mathfrak{X}}.$$

For a in B, we have the inequality

$$\begin{aligned} 2\|a\|_{\mathfrak{X}} &= \|(z + a) - (z - a)\|_{\mathfrak{X}} \\ &\le \|z + a\|_{\mathfrak{X}} + \|z - a\|_{\mathfrak{X}}. \end{aligned} \tag{23}$$

The result follows by taking the supremum over B in (23):

$$\sup_{a\in B} \|a\|_{\mathfrak{X}} \le \sup_{a\in B} \|z - a\|_{\mathfrak{X}} \qquad \forall z \in \mathfrak{X}.$$

It is easy to see that a Chebyshev center for a translate of B, $u + B$, is u. Furthermore, if U: $\mathfrak{X} \to \mathfrak{Z}$ is a linear map into a second normed linear space, then UB is a balanced set in $\mathfrak{Z}$. It follows that $U(u + B) = Uu + UB$ is a translate of a balanced set, and that Uu is a Chebyshev center for this set. Thus U maps the Chebyshev center of $u + B$ to the Chebyshev center of $U(u + B)$. If $\mathfrak{X}$ is a Hilbert space, then the inequality in (23) is strict: it says that the length of one side of a triangle is strictly less than the sum of the other two sides. Hence 0 is a unique Chebyshev center for B and u is the unique Chebyshev center for $u + B$. Denoting the Chebyshev center of a set M by $c(M)$, we can summarize the results of this paragraph:

$$c(u + B) = u$$

$$c(U(u + B)) = Uc(u + B).$$

We adopt $c(UC(y))$ as our solution to the recovery problem: it remains to find it. The key observation is that $C(y)$ always a translate of a convex balanced set. It then follows that $c(UC(y)) = Uc(C(y))$. To find the best estimate over UB, where U is an arbitrary linear map into a normal linear space, it suffices to find the Chebyshev center of $C(y)$, $c(C(y))$, and then compute $Uc(C(y))$.

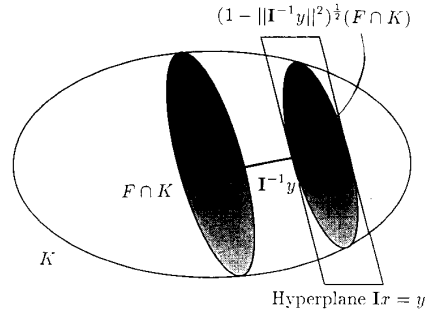

Fig. 8. Hypercircle $C(y)$ an its center, $I^{-1}y$.

C. Chebyshev Center of Hypercircles

The Chebyshev centers of hypercircles are well known in approximation theory: see [3], [8], and [10]. We prove a specialized form of this result, appropriate to our treatment. Fig. 8 may be useful in understanding the theorem.

Theorem 7.1: An alternative characterization for $C(y)$ in (21) is

$$C(y) = I^{-1}y + (1 - \|I^{-1}y\|_{\mathfrak{X}}^2)^{1/2}(F \cap K). \tag{24}$$

In (24), F is the kernel of I. Since $F \cap K$ is convex and balanced, it follows that $c(C(y)) = I^{-1}y$, where $I^{-1}y$ is the element of minimum norm in $\mathfrak{X}$ for which $I(I^{-1}y) = y$.

Proof: Note that $F \cap K$ is convex and balanced because it is the intersection of two convex balanced sets. We assume that $C(y)$ is not empty, otherwise the recovery problem is meaningless. By definition $x_0 = I^{-1}y$ is orthogonal to F. For any signal x is the right-hand side of (24), $Ix = y$ and $\|x\|_{\mathfrak{X}} \le 1$ so the right-hand side is contained in $C(y)$. Conversely, since $C(y)$ is not empty, at least x_0 must be in $C(y)$. For any other signal x in $C(y)$, $x - x_0$ is annihilated by I, so $x - x_0$ lies in F. Hence $x - x_0$ is orthogonal to x_0, and, by the Pythagorean equality,

$$\|x - x_0\|_{\mathfrak{X}}^2 = \|x\|_{\mathfrak{X}}^2 - \|x_0\|_{\mathfrak{X}}^2 \le 1 - \|x_0\|_{\mathfrak{X}}^2.$$

Hence $C(y)$ is contained in the right-hand side of (24), proving the theorem.

VIII. Calculation of the Chebyshev Center

A. Linearity of the Interpolation Filter

The Chebyshev center has been shown to be the minimum $\mathfrak{X}$-norm signal on the hyperplane determined by the known samples. The solution to this problem is well known: see Golomb and Weinberger [3]. If the collection of known samples is y, the minimum norm signal is $I^{-1}y$ where I^{-1} is the pseudoinverse with respect to the $\mathfrak{X}$-norm. The signal $I^{-1}y$ is the unique signal in $\mathfrak{X}$ with

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the property

$$\|\boldsymbol{I}^{-1}y\|_{\mathfrak{X}} = \inf_{\boldsymbol{I}x=y} \|x\|_{\mathfrak{X}}.$$

Since $\boldsymbol{I}^{-1}y$ is of minimum norm, it must be orthogonal to the null space of $\boldsymbol{I}$, F. The subspace, F, is the set of signals which are zero on the time-indices $n = \lambda r$, $\lambda \in \mathbf{Z}$. This is the intersection of the subspaces orthogonal to the representers $\phi_{\lambda r}$, $\lambda \in \mathbf{Z}$:

$$F = \bigcap_{\lambda \in \mathbf{Z}} F_{\lambda r}.$$

The orthogonal complement of F is the subspace spanned by the representers, $\{\phi_{\lambda r}\}_{\lambda \in \mathbf{Z}}$. Since $\boldsymbol{I}^{-1}y$ is orthogonal to F, it lies in this subspace. Hence $\boldsymbol{I}^{-1}y$ is a linear combination of the representers:

$$\boldsymbol{I}^{-1}y = \sum_{\nu} w_\nu \phi_{\nu r}.$$

Insisting that this linear combination goes through the known sample points leads us to solve for the unknown coefficients w_ν:

$$y_{\mu r} = \sum_{\nu \in \mathbf{Z}} w_\nu (\phi_{\mu r}, \phi_{\nu r})_{\mathfrak{X}}.$$

Having solved for these coefficients, we can express the nth sample of $\boldsymbol{I}^{-1}y$ as a linear combination of the known samples, $\{y_{\lambda r}\}_{\lambda \in \mathbf{Z}}$:

$$(\boldsymbol{I}^{-1}y)_n = \sum_{\lambda \in \mathbf{Z}} a_{n\lambda} y_{\lambda r}. \tag{25}$$

We can view (25) as expressing the input–output relationship of a linear time-varying filter, taking the decimated sequence y as its input and producing interpolating sequence, $\boldsymbol{I}^{-1}y$, as its output. The linearity is a consequence of the fact that $\boldsymbol{I}^{-1}$ is a linear operator on the sample space $\mathfrak{Y}$.

B. Time Invariance

We have shown that the optimal recovery operator $\boldsymbol{A}$, can be chosen as the linear operator $\boldsymbol{I}^{-1}$ for the stated interpolation problem. If we have infinite data, then we can show that the optimal interpolation filter is time-invariant. If, as in practice, we only have a finite number of sample points, then the optimal filter is not time invariant. The case of finite-record filtering was analyzed in [11].

To prove time invariance, we examine the interaction of the shift operators S_n with sampling. The shift operator on l_2 is

$$(S_m x)_n = x_{n-m} \qquad x \in l_2.$$

The family of shifts is closed under inversion and composition, and preserves l_2-norm:

$$(S_n)^{-1} = S_{-n}$$

$$S_n S_m = S_{n+m}$$

$$\|S_n x\|_2 = \|x\|_2.$$

Similarly we can define shifts for $\mathfrak{X}$. We use the same symbol S to denote a shift on $\mathfrak{X}$: the context should make clear which shift operator we are discussing. Note that S_n preserves the $\mathfrak{X}$-norm as well.

The range of the sampling operator, $\boldsymbol{I}$, is contained in a subspace of l_2, namely $l_2(T)$. The subspace $l_2(T)$ is not invariant under the family of shifts: for example it is not invariant under S_1. Shifting a sample sequence by one gives a sequence which is nonzero off T, and which is therefore not a sample sequence. However, $l_2(T)$ is invariant under the subfamily $\{S_{\lambda r}\}_{\lambda \in \mathbf{Z}}$: these are the permissible time shifts. Shifting a signal x in $\mathfrak{X}$, by λr shifts the samples by λr:

$$\boldsymbol{I}(S_{\lambda r}x) = S_{\lambda r}(\boldsymbol{I}x). \tag{26}$$

By saying that $\boldsymbol{A}$ is time invariant on $l_2(T)$, we mean that shifting a sample sequence, y in $l_2(T)$, by λr shifts $\boldsymbol{A}y$ by λr:

$$\boldsymbol{A}(S_{\lambda r}y) = S_{\lambda r}(\boldsymbol{A}y).$$

Alternatively, remembering that $c(C(y)) = \boldsymbol{A}y$, we must show that

$$c(C(S_{\lambda r}y)) = S_{\lambda r}(c(C(y))).$$

Time invariance is apparent, once we observe that $C(S_{\lambda r}y) = S_{\lambda r}C(y)$: this follows from the invertibility and norm-preserving properties of shifts, together with (26). It is then straightforward to show that the Chebyshev center of $S_{\lambda r}C(y)$ is $S_{\lambda r}c(C(y))$: again one uses the invertibility and norm-preserving properties of shifts. Note that we have not used the inner product properties of $\mathfrak{X}$ nor have we used the linearity $\boldsymbol{I}^{-1}$. The conclusion of time invariance on $l_2(T)$ holds even if $\mathfrak{A}$ is nonlinear on $l_2(T)$.

We have shown that the $\boldsymbol{A}$ is a time-invariant operator on $l_2(T)$. However, we have taken the sample space, $\mathfrak{Y}$ to be l_2. The question arises as to whether $\boldsymbol{A}$ is time invariant on l_2. By time invariance on l_2, we mean that $\boldsymbol{A}$ intertwines the shift operator on $\mathfrak{X}$ and l_2:

$$\boldsymbol{A}(S_n y) = S_n(\boldsymbol{A}y) \qquad \forall n.$$

So far we have only defined $\boldsymbol{A}$ on the range of $\boldsymbol{I}$, ran $(\boldsymbol{I})$: it agrees with $\boldsymbol{I}^{-1}$ on ran $(\boldsymbol{I})$. We are free to extend $\boldsymbol{A}$ to the rest of l_2, as long as the restriction of this extended operator agrees with $\boldsymbol{I}^{-1}$ on ran $(\boldsymbol{I})$. One natural candidate is to take the pseudoinverse $\boldsymbol{I}^{-1}$.

The pseudoinverse of $\boldsymbol{I}$, $\boldsymbol{I}^{-1}$ is first defined on the range of $\boldsymbol{I}$, ran $(\boldsymbol{I})$. It is then extended to the rest of l_2 by defining $\boldsymbol{I}^{-1}$ on the orthogonal complement of the range of $\boldsymbol{I}$, ran $(\boldsymbol{I})^{\perp}$: by linearity, $\boldsymbol{I}^{-1}$ is well defined on all of $\mathfrak{Y}$. The convention is to require $\boldsymbol{I}^{-1}$ to annihilate ran $(\boldsymbol{I})^{\perp}$: this is the "least squares, minimum norm" inverse. For any signal y in l_2, which is zero on T, it follows that $\boldsymbol{I}^{-1}y = 0$. In particular, let y be a sample sequence in the range of $\boldsymbol{I}$, and let $S_1 y$ be the same sequence shifted by a one time index:

$$(S_1 y)_n = y_{n=1}.$$

Then if $\boldsymbol{I}^{-1}$ is the pseudoinverse, we have $\boldsymbol{I}^{-1}(S_1 y) = 0$.

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Clearly, the pseudoinverse is not a time-invariant operator on l_2. However, we can define $\boldsymbol{A}$ so that it is time invariant on l_2 and so that its restriction to $l_2(T)$ is equal to the restriction of $\boldsymbol{I}^{-1}$ to $l_2(T)$.

Observe that l_2 is the direct sum of the orthogonal subspace $S_n l_2(T)$, $n = 0, \cdots, r-1$:

$$l_2 = \bigoplus_{n=0}^{r-1} S_n l_2(T).$$

If y is in $l_2(S_n(T))$, we define $\boldsymbol{A}y$ as $S_n \boldsymbol{A} S_{-n}\, y$ This extends $\boldsymbol{A}$ to a linear operator which is time invariant on l_2 and which agrees with $\boldsymbol{I}^{-1}$ on $l_2(T)$.

Physically, we can use a time-invariant filter which takes the decimated sequence at the low rate and produces the interpolated signal at the high rate. Alternatively, we can zero pad the decimated signal to make it high rate, and then use a time-invariant filter to produce the interpolated signal.

C. Equivalence of Estimation and Approximation

In this subsection we shall show that the estimation of a missing sample is equivalent to approximating the representer of the missing sample by a linear combination of the representers of the known samples. We shall use this observation to pose the interpolation problem as a filter design in the frequency domain. This result has been observed in [12].

We remarked earlier that the recovery of an individual sample can be viewed as estimation of that sample by a linear combination of the known samples. We seek an estimate, $(\widehat{x_n})$, for x_n, which is a linear combination of the known samples, y_λ:

$$(\widehat{x_n}) = \sum_{\lambda \in \mathbf{Z}} a_{n\lambda}\, y_\lambda.$$

We form the error functional

$$E(x) = x_n - \sum_{\lambda \in \mathbf{Z}} a_{n\lambda}\, y_\lambda.$$

The representer for the functional E is $\phi_n - \Sigma_{\lambda \in \mathbf{Z}}\, a_{n\lambda}\phi_{\lambda r}$. The image of $C(y)$ under E is an interval, centered around $E(c(C(y)) = E\boldsymbol{I}^{-1}y)$. This is shown schematically in Fig. 9. The length of this interval is independent of the coefficients, $\{a_{n\lambda}\}$. This can be seen, by using (24):

$$E(x) = ((\phi_n - \sum_{\lambda \in \mathbf{Z}} a_{n\lambda}\phi_{\lambda r}), \boldsymbol{I}^{-1}y) + (\phi_n, z)$$

$$z \in (1 - |\boldsymbol{I}^{-1}y\|^2)^{1/2} F \cap K. \qquad (27)$$

We have used the fact that F is orthogonal to the space spanned by the representers $\{\phi_{\lambda r}\}_{\lambda \in \mathbf{Z}}$ in (27). As z ranges over $L(y) = (1 - |\boldsymbol{I}^{-1}y\|^2)^{1/2} F \cap K$, $E(x)$ varies around the point $E(\boldsymbol{I}^{-1}y)$. If P_F is the projection onto the subspace, F, in the $\mathfrak{X}$-norm, then the length of $E(C(y))$ is

$$2 \sup_{z \in L(y)} |(\phi_n, z)_{\mathfrak{X}}| = 2(1 - \|\boldsymbol{I}^{-1}y\|^2)^{1/2} \|P_F \phi_n\|_{\mathfrak{X}}.$$

The worst case error is the end point of $E(C(y))$ which is further away from the origin. To minimize the worst case error, we need to arrange for $E(c(C(y)))$ to be as close to the origin as possible. In fact we can make $E(c(C(y)))$ zero by a suitable choice of $\{a_{n\lambda}\}$. Since $c(C(y)) = \boldsymbol{I}^{-1}y$ is a linear combination of the representers, it suffices to select coefficients $\{a_{n\lambda}\}$ to make $\phi_n - \Sigma_{\lambda \in \mathbf{Z}}\, a_{n\lambda}\phi_{\lambda r}$ orthogonal to the space spanned by the representers. By the orthogonality principle, this is just the best approximation of ϕ_n by a linear combination of the representers $\phi_{\lambda r}$. Hence the coefficient, $\{a_{n\lambda}\}$, in (25) are the same coefficients which make $\Sigma_\lambda a_{n\lambda}\phi_{\lambda r}$ the best linear approximation to ϕ_n.

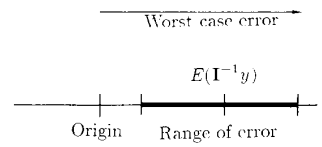

Fig. 9. The range of values of the error functional.

Since the estimation problem is an approximation problem, we observe that the recovery of a sample is related to the filter-design problem: note that (15) has precisely the form of an approximation problem.

IX. Conclusions

We have examined the interpolation problem from the viewpoint of filter design and in the context of optimal recovery. The work described in [1] was briefly reviewed. It was then shown that minimizing the L_2 norm of an appropriate error frequency response, as was done in [1], gives a filter which interpolates the known samples: this constraint need not be incorporated into the filter-design problem beforehand. The minimization of the L_∞-norm (Chebyshev norm) of the error frequency response does not give an interpolation filter in general.

We then described the elements of deterministic estimation theory, and of optimal recovery. This lent some useful insight into how the geometry of the filter class affects the best estimate for a missing sample. In particular, the hypercircle $C(y)$ is a translate of a balanced set. Since linear transformations preserve this property and map Chebyshev centers to Chebyshev centers, it suffices to compute the Chebyshev center of $C(y)$ for any optimal recovery problem. In other words, we showed that

$$c(\boldsymbol{U}C(y)) = \boldsymbol{U}c(C(y)).$$

This explains why optimal recovery of an individual sample is equivalent to optimal recovery of the whole signal. We note that there are other signal classes of interest, such as the class of amplitude bounded signals dealt with in [2], which do not have such nice geometries.

The best interpolation filter for our stated problem is linear, although this is not assumed in the optimal recovery formulation of the problem. If we have an infinite number of equispaced samples, then the filter is also time invariant. The intersection of the hyperplane, determined by the known samples, with the filter class ball has the property that its Chebyshev center is a linear transfor-

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mation of the known sample: it is the minimum norm signal determined by the samples, as given by the Golomb–Weinberger theory. The time invariance of the filter is related to the fact that time shifts preserve the weighted l_2 norms that we use, and to the interplay between sampling and time shifts.

The problem of estimating a linear functional from known linear functionals is equivalent to one of approximating the representer of the unknown functional by a linear combination of the representers of the known functionals. From this viewpoint, the design of the interpolation filter is equivalent to minimization of an integrated squared error in the frequency domain. This explains why one may start out by designing a filter in the frequency domain and obtain an interpolation filter as a design solution.

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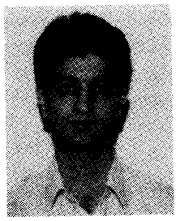

Ram G. Shenoy (M'92) received the M.A. degree in electrical sciences from Cambridge University, U.K., and the Ph.D. degree from Cornell University.

He is now with Schlumberger-Doll Research, Ridgefield, CT. His interests are in signal processing, particularly multirate filtering, multiresolution analysis and interpolation.

Thomas W. Parks (S'66–M'67–SM'79–F'82) received the Ph.D. degree from Cornell University in 1967.

He joined Rice University, Houston, TX, where he was a Professor of Electrical Engineering until he joined the School of Electrical Engineering at Cornell University in 1986. He is engaged in research on signal theory and digital signal processing. His research interests are in the areas of time-frequency and wavelet analysis, signal reconstruction, array processing for sonar and seismic applications, digital filter design, pattern classification, and neural networks. He serves as a consultant to industry.

Dr. Parks received the Alexander von Humboldt Senior Scientist Award. He has also received the Society Award and the Technical Achievement Award of the IEEE Acoustics, Speech, and Signal Processing Society. He has been an Associate Editor for the IEEE Transactions on Acoustics, Speech, and Signal Processing and a member of the Administrative Committee of that society.

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